Does How People Feel Predict How They Default?
A Sentiment-Driven Credit Risk Model Using Macroeconomic Indicators

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ABSTRACT

This paper investigates whether consumer sentiment contains predictive information for credit card delinquency beyond unemployment, inflation and interest rates. Using quarterly US data from 1980 to 2024, we estimate a Vector Autoregression (VAR), run bivariate Granger causality tests, and fit logistic regression classifiers of high-delinquency quarters. Sentiment does not Granger-cause delinquency at lags 1 to 4. Adding lagged sentiment to the logistic model raises in-sample accuracy from 82.8% to 85.8%, a modest difference that we do not formally test. We also apply the Basel-style Expected Loss formula (EL = PD x LGD x EAD) as an illustration of how predicted probabilities translate into loss estimates. This is a preliminary exercise; out-of-sample evaluation is left to future work.

1. Introduction

Traditional credit risk models rely on macroeconomic fundamentals such as unemployment, inflation and interest rates. A literature in behavioural economics suggests that survey measures of consumer confidence may contain forward-looking information not captured by such data. We ask whether the Michigan Consumer Sentiment Index improves the prediction of credit card delinquency when added to a standard set of macroeconomic controls.

The question matters for financial stability and for risk management: the Basel framework requires institutions to model Probability of Default (PD), Loss Given Default (LGD) and Exposure at Default (EAD), and forward-looking indicators may be useful in macroprudential stress testing.

2. Data

All data come from the Federal Reserve Bank of St. Louis (FRED) API, 1980 Q1 to 2024 Q4:

Variable	FRED code	Description

Consumer sentiment	UMCSENT	Michigan Consumer Sentiment Index
Delinquency rate	DRCCLACBS	Credit card delinquency rate, commercial banks, 30+ days past due (%)
Unemployment	UNRATE	Civilian unemployment rate (%)
CPI	CPIAUCSL	Consumer Price Index, all urban
Fed funds rate	FEDFUNDS	Effective federal funds rate (%)

Table 1: Data sources.

Monthly series are averaged to quarterly frequency. After lagging and removing missing values, the classification sample has 134 quarters and includes several recessions (1990-91, 2001, 2008-09, 2020).

3. Methodology

3.1 Stationarity

Augmented Dickey-Fuller tests find sentiment, delinquency and CPI non-stationary in levels ($p > 0.05$); unemployment and the federal funds rate test as stationary. All series are first-differenced before VAR estimation.

3.2 VAR and Granger causality

We estimate a VAR on the differenced series with lag order chosen by AIC (two lags). We then run bivariate Granger causality tests on differenced delinquency and sentiment at lags 1 to 4. These tests do not control for the other variables.

3.3 Logistic regression

A quarter is labelled high risk if delinquency exceeds its sample median (3.91%), so the two classes are balanced by construction. A baseline model uses one-quarter lags of unemployment, CPI and the federal funds rate (in levels, standardised); an augmented model adds one- and two-quarter lags of sentiment. Accuracy is computed in-sample.

3.4 Expected loss

Expected Loss is computed as:

$$\mathbf{EL = PD \times LGD \times EAD}$$

PD is the fitted probability from the augmented logistic model. LGD is set to an illustrative 45%, the Basel foundation-IRB supervisory value for senior unsecured corporate exposures (Basel does not set a supervisory LGD for credit cards). EAD is normalised to one.

4. Results

4.1 VAR and Granger tests

Lagged delinquency is the strongest predictor of delinquency in the VAR ($p < 0.001$). Lagged unemployment (second lag, $p = 0.021$) and lagged CPI (second lag, $p = 0.019$) are also significant, while sentiment is not. The bivariate Granger tests find no effect of sentiment on delinquency at any lag (all $p > 0.38$).

4.2 Classification

The baseline model achieves 82.8% in-sample accuracy and the augmented model 85.8%, a gain of 3.0 percentage points. For the augmented model, precision for high-risk quarters is 83% and recall is 89%.

Model	Accuracy (in-sample)	Precision (high risk)	Recall (high risk)
Baseline (macro only)	82.8%	not computed	not computed
Augmented (+ sentiment)	85.8%	83%	89%

Table 2: Model comparison.

4.3 Expected loss illustration

Average PD is about 49% and average EL 22.2%. Because the label is defined relative to the median, average PD is close to 50% by construction, so the level of EL is illustrative rather than a loss estimate. The variation over time is more informative: the five highest-EL quarters all fall in 1991-92, the early-1990s recession.

5. Discussion and Limitations

Sentiment does not Granger-cause delinquency, yet it raises in-sample classification accuracy slightly. These results are not contradictory, since the tests differ (bivariate differenced series versus a levels-based classifier), but they cannot be compared directly, and the improvement may reflect non-linearity or noise.

Main limitations: (i) accuracy is in-sample and the 3-point gain is not tested for significance; (ii) the median-based label and trending delinquency may let the model partly capture time effects; (iii) Granger tests are bivariate; (iv) national aggregates hide heterogeneity; (v) LGD is fixed and illustrative. Natural extensions are a time-based train/test design, modelling the continuous delinquency rate, non-linear models and time-varying LGD.

6. Conclusion

This exercise finds, at best, suggestive evidence that sentiment carries a small amount of extra information about high-delinquency periods. The evidence is preliminary: it is in-sample, untested for significance, and sensitive to how risk is defined. The Expected Loss step illustrates the Basel formula but does not provide a reliable loss estimate.

References

Basel Committee on Banking Supervision (2017). Basel III: Finalising post-crisis reforms. BIS.

Carroll, C. D., Fuhrer, J. C., & Wilcox, D. W. (1994). Does consumer sentiment forecast household spending? American Economic Review, 84(5), 1397-1408.

Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. Econometrica, 37(3), 424-438.

Sims, C. A. (1980). Macroeconomics and reality. Econometrica, 48(1), 1-48.

University of Michigan. Surveys of Consumers; Board of Governors of the Federal Reserve System. Data via FRED, Federal Reserve Bank of St. Louis.